Next, see how to interface with an external sensor for capturing real-time temperature and humidity. We'll use an inexpensive DHT 11 sensor for that. MicroPython already includes a module to handle these sensors out-of-the-box. First, assemble the connections as shown in Figure 4 and execute the code given below to read the sensor output (marked as S on the sensor) through GPIO 5 Pico pin 7 ('-' marked on the sensor connected to Pico pin 3 GND and middle VCC to pin 36 3V3(OUT)):

```
--- Start of dht11temphmdt.py ---

import time
import machine
import dht

sensor_pin = machine.Pin(5)
sensor = dht.DHT11(sensor_pin)

print(" Starting DHT11 sensor readings...")
print(" Press Ctrl+C to stop\n")

while True:
    try:
        # DHT11 needs at least 2 seconds between readings
        time.sleep(2)

        # Trigger a measurement
        sensor.measure()

        # Read the values
        temp = sensor.temperature()
        hum = sensor.humidity()


        # Print formatted output
        print(f" Temperature: {temp}°C , Humidity: {hum}%")

    except Exception as e:
        print(f"Read error: {e}")
       # DHT11 Blue Case Sensor: S => Output, - => GND,
middle => 3V3
        print("Check your wiring!\n")

--- End of dht11temphmdt.py ---
```

This should dump the surrounding atmospheric temperature and humidity on your terminal. Just blow your breath into the sensor and you'll be able to see the difference in humidity values. You can also trigger other outputs like driving different colour LEDs based on different temperature ranges, switching external devices like fans on or off, etc, based on your needs and creativity.

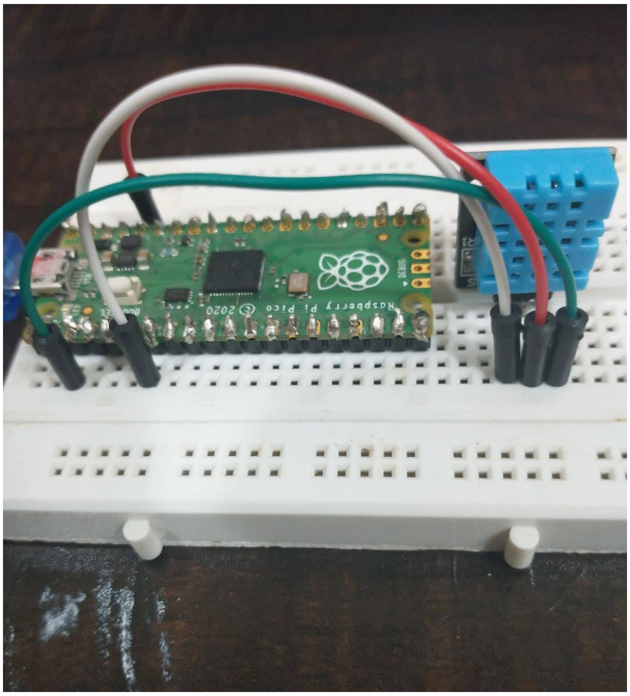

Figure 4: External temperature humidity sensing

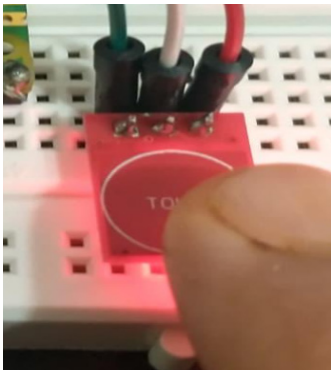
Figure 5: External touch sensing

The next sensor you can interface with your board is the inexpensive TTP 223. This touch sensor outputs a high signal when you bring your finger near it and a low signal when the finger is away. It can also be used as a latch, toggling output at every touch to switch external devices on or off. Wire the TTP 223 to your board as shown in Figure 5 and execute the code given below to read the sensor output (marked as I/O on the sensor) through GPIO 5 Pico pin 7 (GND marked on sensor connected to Pico pin 3 GND and VCC to pin 36 3V3(OUT)) and, accordingly, turn on an external red-green-blue LED pattern:

```
--- Start of ttp223tchctrl.py ---

from machine import Pin
from time import sleep

touch = Pin(5, Pin.IN)
rled = Pin(0, Pin.OUT)
```